

Requirements Specification

Requirement Gathering

Requirements were collected based on the needs of an IT support team aiming to improve ticket assignment efficiency, reduce resolution time, and enhance visibility of support operations within the ServiceNow platform.

The system should automate ticket routing, minimize manual intervention, and ensure fair workload distribution among support agents.

Functional Requirements

User & Role Management

- The system shall allow administrators to create and manage users.
- Roles such as **Support Agent**, **Support Manager**, and **System Administrator** shall be defined.
- Role-based access control (RBAC) shall restrict features based on responsibilities.
- Support groups shall be created to organize agents based on specialization (e.g., Network, Hardware, Software).

Ticket Management

- The system shall use the Incident table to log and manage support tickets.
- Tickets shall include fields such as Category, Subcategory, Priority, Assignment Group, Assigned To, Status, SLA, and Description.
- Users shall be able to create tickets through the Service Portal or platform UI.

Automated Ticket Assignment

- Ticket assignment shall be automated using **Flow Designer** or Assignment Rules.
- Tickets shall be routed based on predefined conditions such as:
 - Category or Subcategory
 - Priority level
 - Location
 - Workload of support agents

- The system shall support round-robin or load-balanced assignment within groups.

SLA & Escalation Management

- SLA policies shall be configured based on ticket priority.
- Automatic escalation shall occur if SLA thresholds are breached.
- Notifications shall be sent to managers during escalations.

Notifications & Alerts

- Email and in-platform notifications shall be sent for:
 - Ticket assignment
 - Status updates
 - SLA warnings
 - Ticket closure

Reporting & Monitoring

- Dashboards shall provide real-time insights into:
 - Open vs Closed tickets
 - SLA compliance rate
 - Agent workload
 - Average resolution time
- Support Managers shall monitor performance using reports and Performance Analytics.

Ticket Updates & Collaboration

- Agents shall update ticket status and add work notes.
- Attachments and comments shall be supported.
- Managers shall reassign tickets if required.

Non-Functional Requirements

- The system must ensure secure access through role-based permissions.
- The system should provide high availability and reliability.
- The user interface should be simple, responsive, and user-friendly.
- The solution should support scalability for increasing ticket volume.
- Performance should ensure minimal delay in ticket assignment and routing.

- The system must comply with ServiceNow platform best practices

Constraints

- Implementation shall be done exclusively on a ServiceNow Developer Instance.
- Development must follow a configuration-based approach using:
 - Incident Table
 - Assignment Rules
 - Flow Designer
 - SLAs
 - Notifications
 - Access Controls
- Custom scripting should be minimized.
- The solution must remain upgrade-compatible with future ServiceNow releases.
- Access shall be restricted to authorized users only

References

<https://developer.servicenow.com>

<https://docs.servicenow.com>

<https://community.servicenow.com>

<https://nowlearning.servicenow.com>